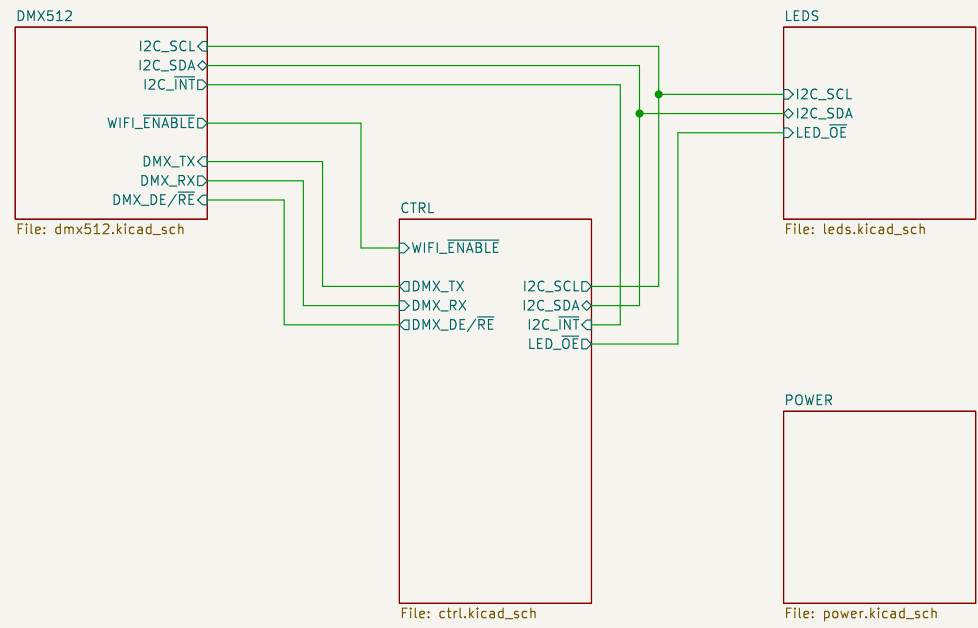


LED Beacon



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

USB Programming

Controller

VCC Monitoring

Temperature Monitoring

Auto reset

Status LED

Does this need to be a switch?
Cheaper switch?
Side actuation?

ROTATE

Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title:

Size: A4 Date: 2026-04-10 Rev: 0

KiCad E.D.A. 9.0.9 Id: 2/5

Controller

USB Programming

J1 USB_C_Receptacle_USB2.0_16P

VBUS A4

CC1 A5

CC2 B5

D- B7

D+ A6

SBU1 A8

SBU2 B8

SHIELD S1

GND A1

R1 5k1

R2 5k1

GNDND

C25 100nF

D5 SS34

VDC

VCC Monitoring

VCC_MON

R6 62k

R7 10k

C4 100nF

GND

Voltage Monitor

Vmax is 17.5V

$V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$

$V_{adc} \times 5.3 = VCC$

1lsb = 4,271μV

Temperature Monitoring

TEMP_MON

R3 10k

TH1 NCP18XH103F03RB

C3 1uF 50V

GND

Auto reset

+3V3

R41 10k

ESP_RST

D6 BAT54WS

ESP_BOOT

C26 100nF

C27 100uF

GND

SW2

SW_Push

GND

Status LED

R40 R

LED1 KT-0603R

GND

Wires & Bits

Sheet: /CTRL/

File: ctrl.kicad_sch

Title:

Size: A4

Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

Id: 2/5

USB Programming

Controller

VCC Monitoring

Temperature Monitoring

Auto reset

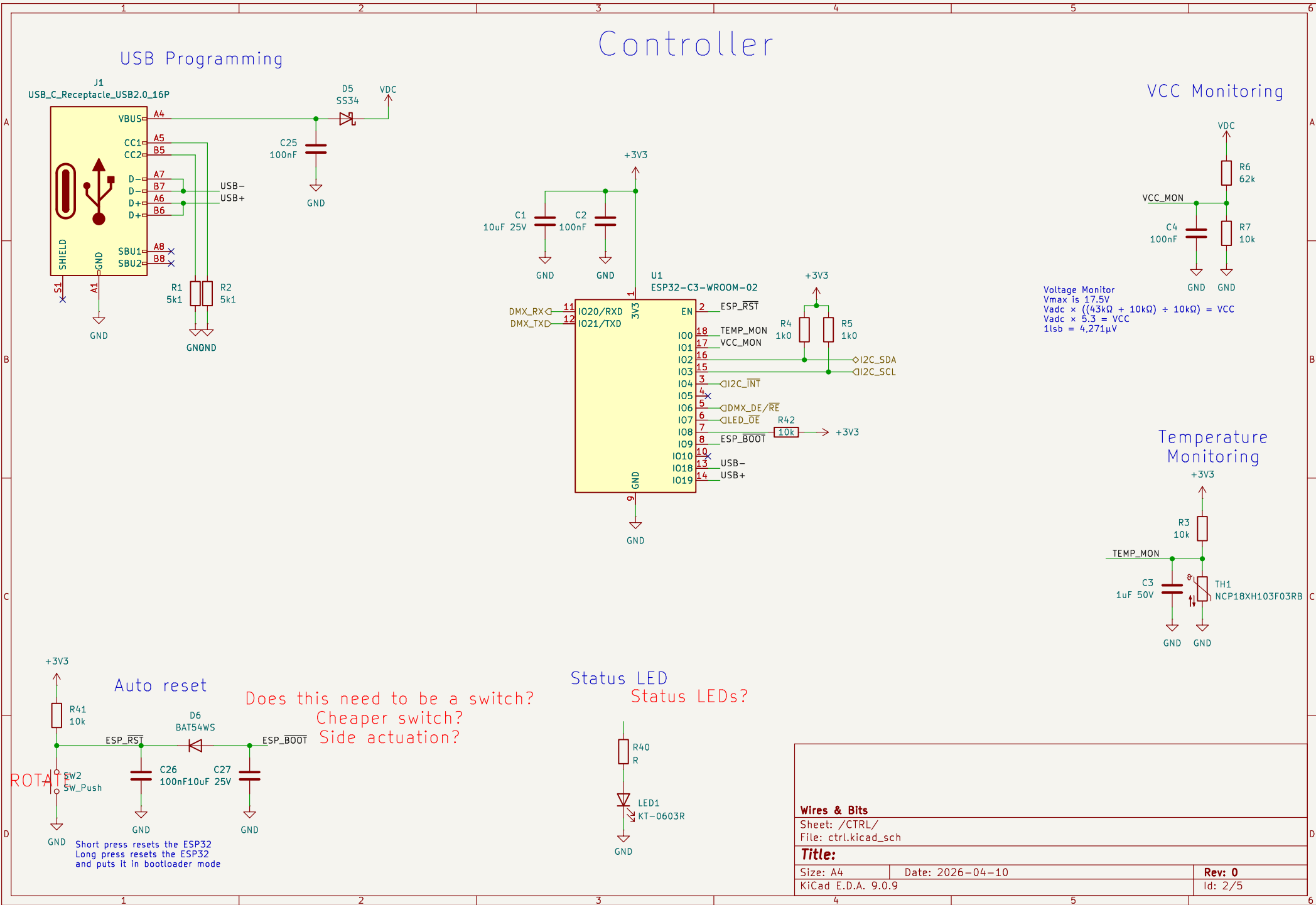
Status LED

Does this need to be a switch?
Cheaper switch?
Side actuation?

ROTATE

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Wires & Bits		
Sheet: /CTRL/		
File: ctrl.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 2/5



USB Programming

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Temperature Monitoring

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Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 2/5

USB Programming

Controller

VCC Monitoring

Temperature Monitoring

Auto reset

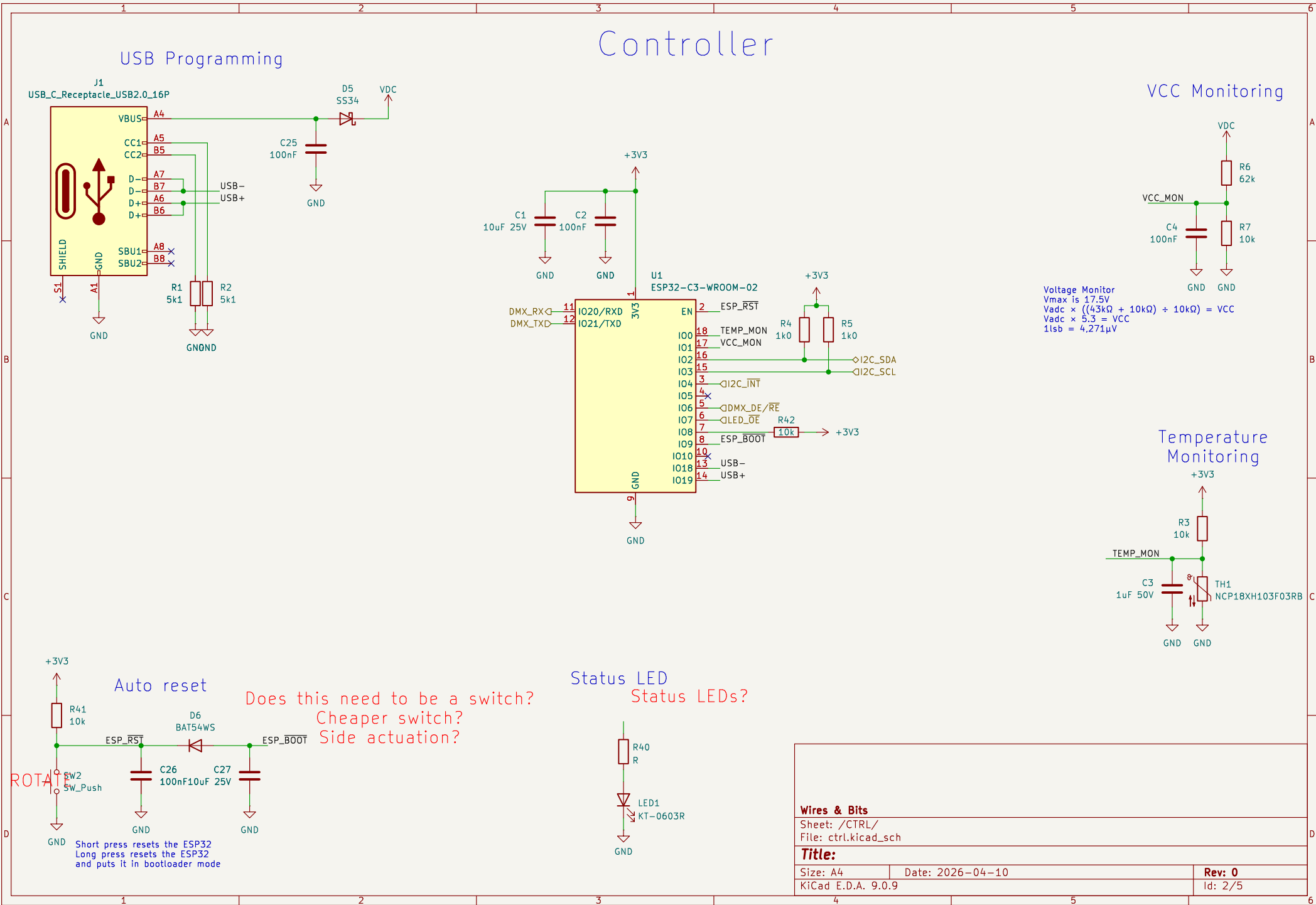
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Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 2/5



Controller

USB Programming

J1 USB_C_Receptacle_USB2.0_16P

VBUS A4

CC1 A5

CC2 B5

D- B7

D+ A6

SBU1 A8

SBU2 B8

D5 SS34

C25 100nF

GND

VDC

R1 5k1

R2 5k1

GND

GND

VCC Monitoring

VCC_MON

R6 62k

R7 10k

C4 100nF

GND

GND

VDC

Voltage Monitor

Vmax is 17.5V

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1lsb = 4,271μV

Temperature Monitoring

TEMP_MON

R3 10k

TH1 NCP18XH103F03RB

C3 1uF 50V

GND

GND

Status LED

R40 R

LED1 KT-0603R

GND

Auto reset

R41 10k

D6 BAT54WS

ESP_RST

C26 100nF

C27 100nF

GND

GND

GND

SW2 SW_Push

ROTATE

Short press resets the ESP32

Long press resets the ESP32 and puts it in bootloader mode

Does this need to be a switch?

Cheaper switch?

Side actuation?

Wires & Bits

Sheet: /CTRL/

File: ctrl.kicad_sch

Title:

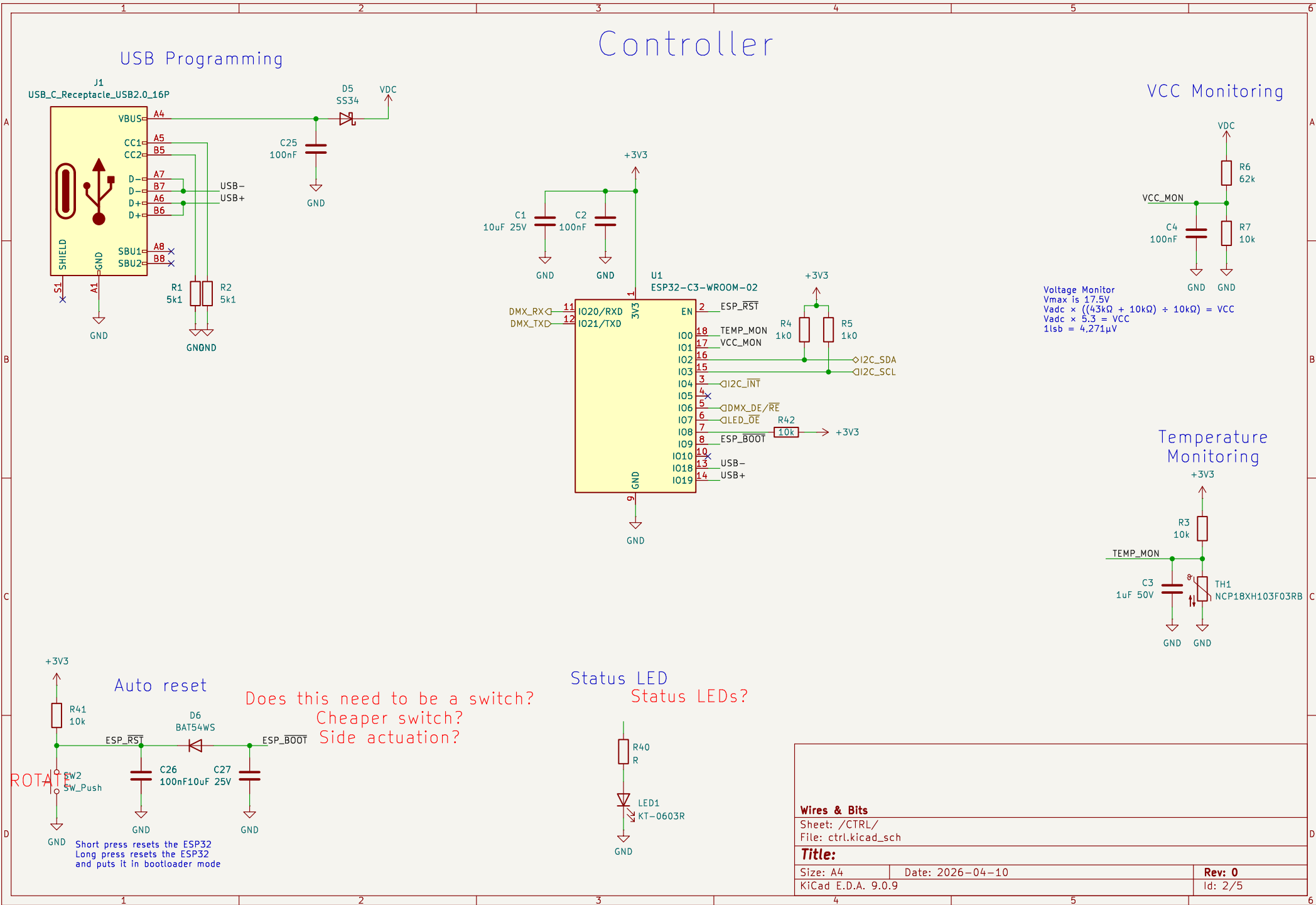
Size: A4

Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

Id: 2/5



Controller

USB Programming

J1 USB_C_Receptacle_USB2.0_16P

VBUS A4

CC1 A5

CC2 B5

D- B7

D+ A6

SBU1 A8

SBU2 B8

SHIELD S1

GND A1

R1 5k1

R2 5k1

GNDND

C25 100nF

D5 SS34

VDC

VCC Monitoring

VCC_MON

R6 62k

R7 10k

C4 100nF

GND

Voltage Monitor

Vmax is 17.5V

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1lsb = 4,271μV

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R3 10k

TH1 NCP18XH103F03RB

C3 1uF 50V

GND

Status LED

Status LEDs?

R40 R

LED1 KT-0603R

GND

Auto reset

Does this need to be a switch?

Cheaper switch?

Side actuation?

ESP_RST

D6 BAT54WS

ESP_BOOT

C26 100nF

C27 10uF 25V

GND

R41 10k

SW2

SW_Push

GND

Short press resets the ESP32

Long press resets the ESP32 and puts it in bootloader mode

Wires & Bits

Sheet: /CTRL/

File: ctrl.kicad_sch

Title:

Size: A4

Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

Id: 2/5

USB Programming

J1
USB_C_Receptacle_USB2.0_16P

VBUS A4
CC1 A5
CC2 B5
D- B7
D+ A6
D+ B6
SBU1 A8
SBU2 B8
SHIELD S1
A1
GND
R1 5k1
R2 5k1
GND
GND

D5
SS34
VDC
C25
100nF
GND

Controller

U1
ESP32-C3-WROOM-02

EN 2 ESP_RST
IO0 18
IO1 17 VCC_MON
IO2 16
IO3 15 I2C_SDA
IO4 3 I2C_INT
IO5 4
IO6 5 DMX_DE/RE
IO7 6 LED_OE
IO8 7
IO9 8 ESP_BOOT
IO10 10
IO18 13 USB-
IO19 14 USB+
GND 9
GND

C1 10uF 25V
C2 100nF
GND
GND
+3V3
R4 1k0
R5 1k0
+3V3
R42 10k
+3V3

VCC Monitoring

VCC_MON
C4 100nF
GND
GND
R6 62k
R7 10k
VDC

Voltage Monitor
Vmax is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
1lsb = 4,271μV

Temperature Monitoring

TEMP_MON
C3 1uF 50V
GND
GND
R3 10k
TH1 NCP18XH103F03RB
+3V3

Auto reset

R41 10k
+3V3
ESP_RST
D6 BAT54WS
ESP_BOOT
C26 100nF
C27 10uF 25V
GND
GND
GND

Does this need to be a switch?
Cheaper switch?
Side actuation?

ROTATE

SW2
SW_Push
GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LED

R40 R
LED1 KT-0603R
GND

Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title:

Size: A4
Date: 2026-04-10
KiCad E.D.A. 9.0.9

Rev: 0
Id: 2/5

Controller

USB Programming

J1 USB_C_Receptacle_USB2.0_16P

VBUS A4

CC1 A5

CC2 B5

D- B7

D+ A6

SBU1 A8

SBU2 B8

SHIELD S1

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R1 5k1

R2 5k1

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C25 100nF

D5 SS34

VDC

VCC Monitoring

VCC_MON

R6 62k

R7 10k

C4 100nF

GND

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TEMP_MON

R3 10k

TH1 NCP18XH103F03RB

C3 1uF 50V

GND

Status LED

Status LEDs?

R40 R

LED1 KT-0603R

GND

Auto reset

Does this need to be a switch?

Cheaper switch?

Side actuation?

ESP_RST

D6 BAT54WS

ESP_BOOT

C26 100nF

C27 10uF

GND

R41 10k

SW2

SW_Push

GND

Short press resets the ESP32

Long press resets the ESP32 and puts it in bootloader mode

Wires & Bits

Sheet: /CTRL/

File: ctrl.kicad_sch

Title:

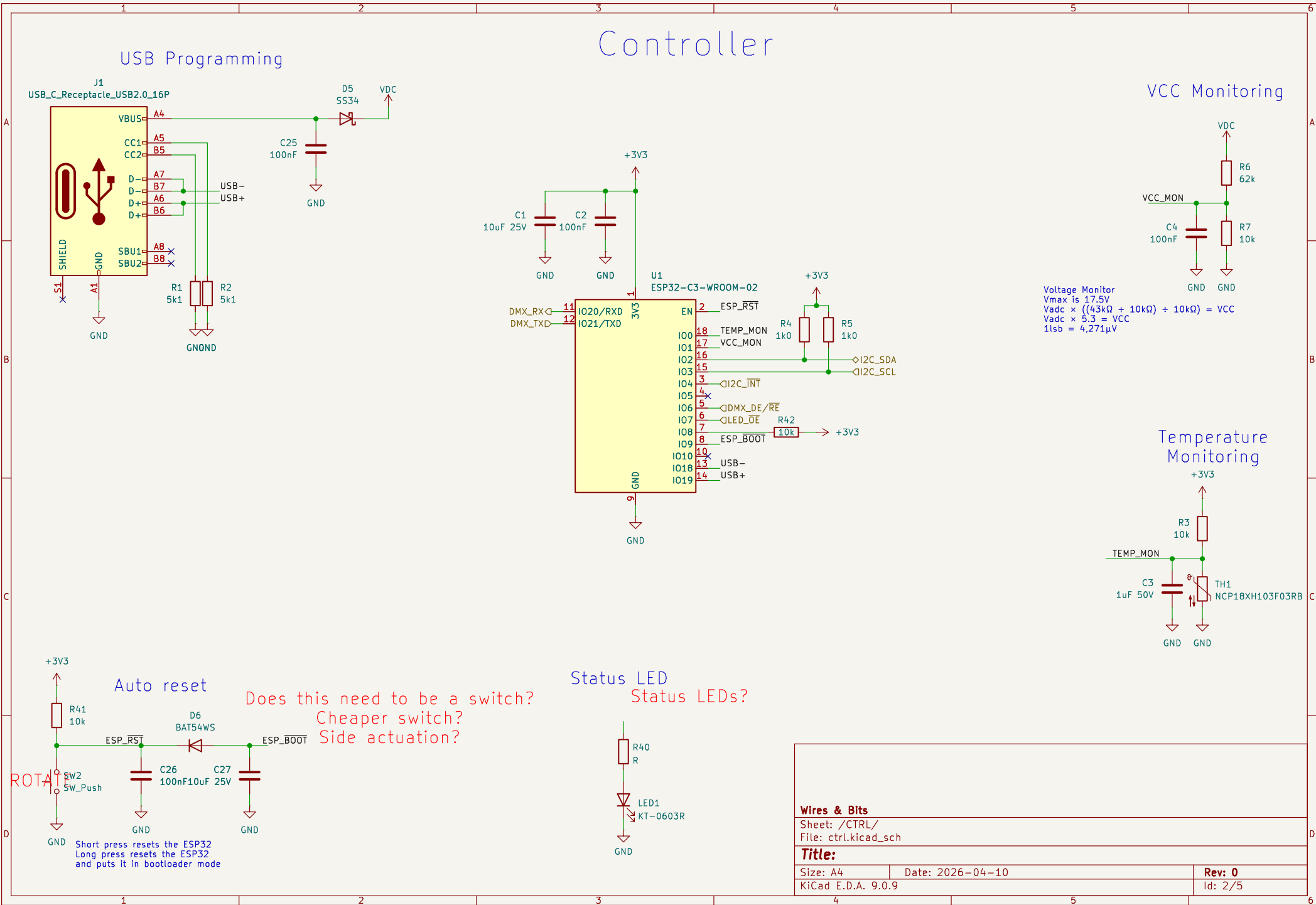
Size: A4

Date: 2026-04-10

Rev: 0

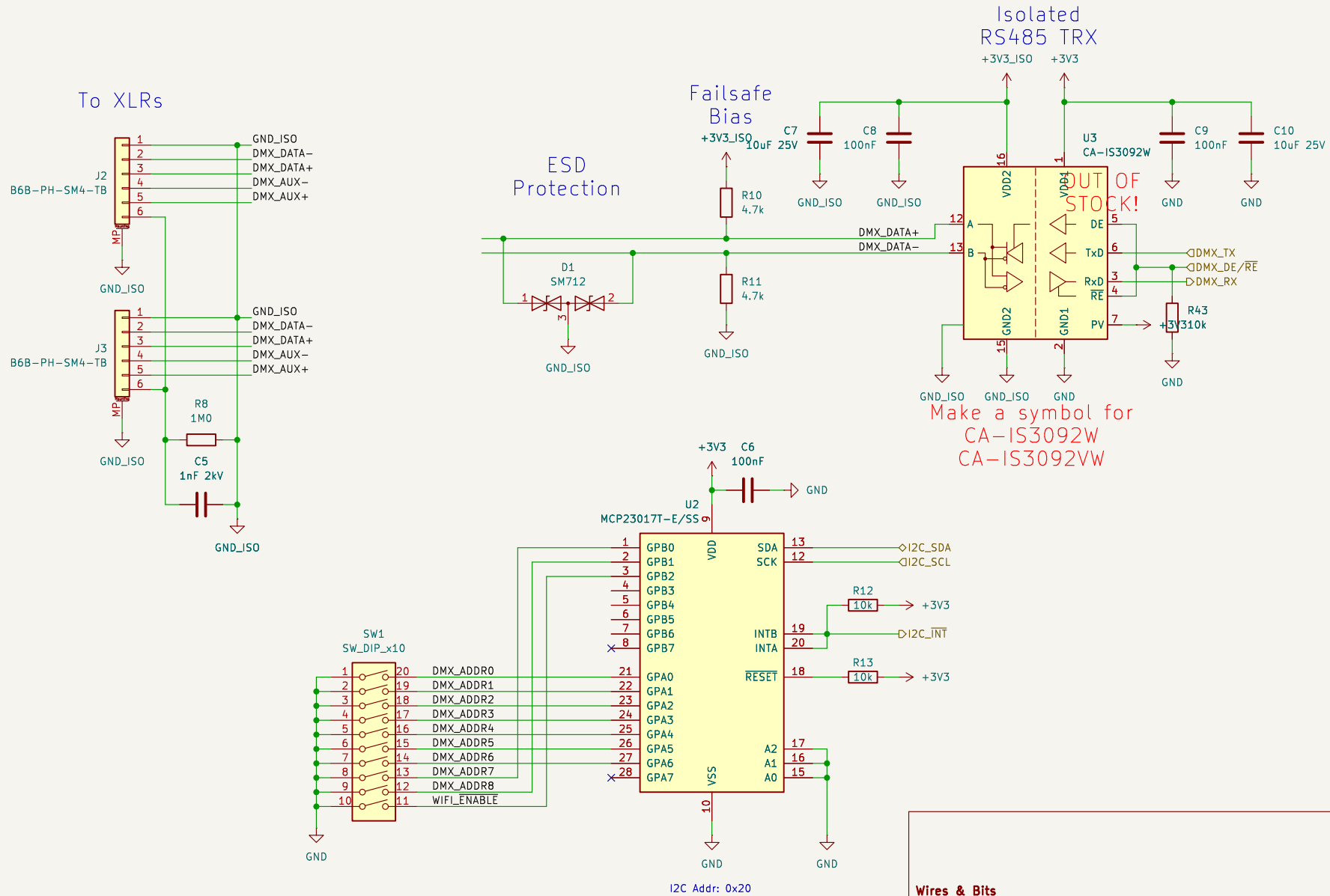
KiCad E.D.A. 9.0.9

Id: 2/5



Id: 2/5

DMX512



Wires & Bits

Sheet: /DMX512/

File: dmx512.kicad_sch

Title:

Size: A4

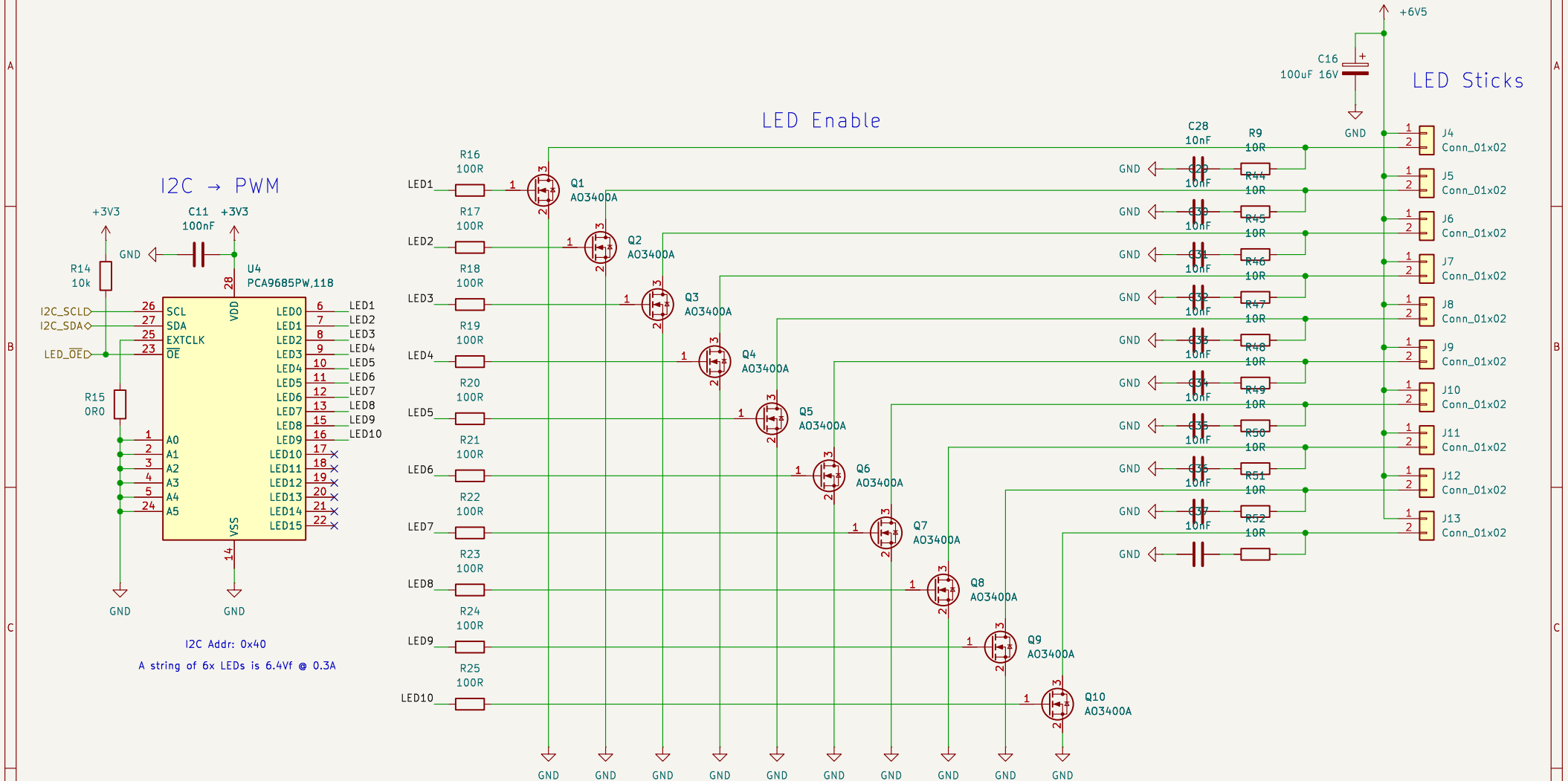
Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

Id: 3/5

LED Driver



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

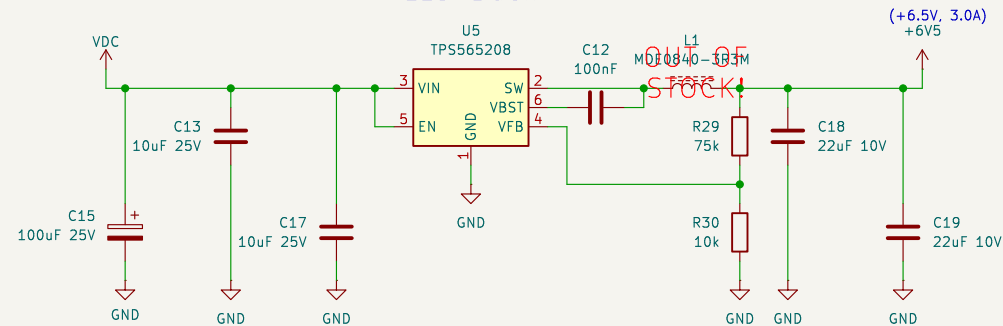
Title:

Size: A4 Date: 2026-04-10
KiCad E.D.A. 9.0.9

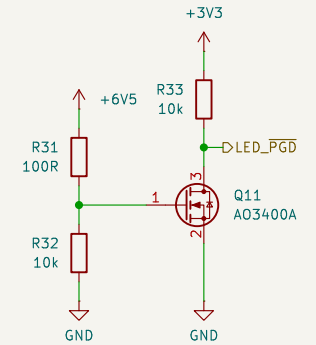
Rev: 0
Id: 4/5

Power Supplies

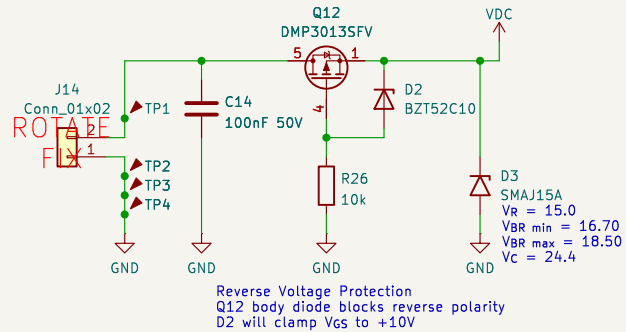
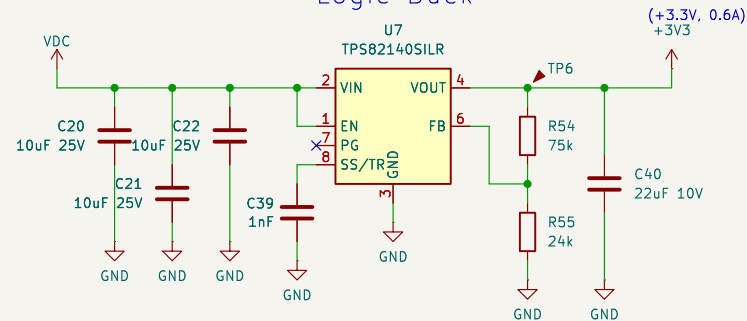
LED Buck



LED Power Good



Logic Buck



Find a good SMD Power input connector
lever terminal blocks?

EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0

Id: 5/5